

ML Expense Analyzer Project Report

Project: ML Expense Analyzer

Version: 1.0.0

Description: Machine Learning project for personal expense analysis and prediction

Generated: 2026-09-28 23:54:04

Dataset Information:

- Total Records: 1,000
- Date Range: 2025-09-28 to 2026-09-28
- Total Expenses: \$189,031.93
- Average Transaction: \$189.03
- Unique Categories: 9

Machine Learning Components:

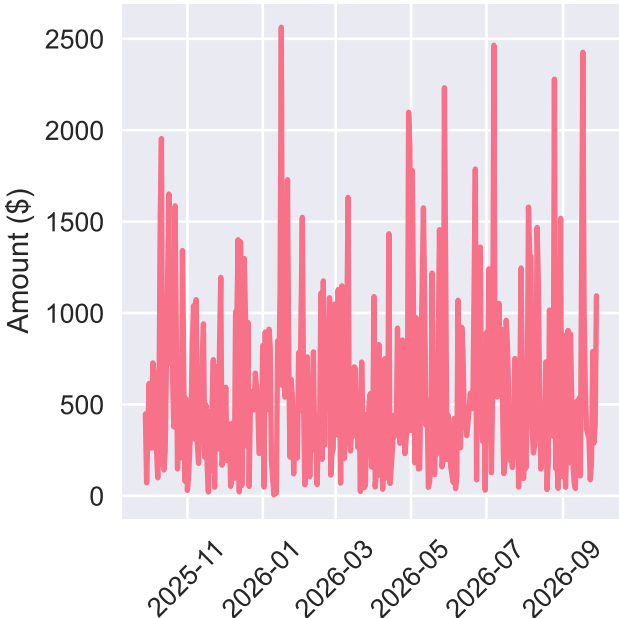
- Supervised Learning: Expense Categorization (Classification)
- Supervised Learning: Future Expense Prediction (Regression)
- Unsupervised Learning: Spending Pattern Discovery (Clustering)
- Anomaly Detection: Fraudulent Transaction Identification

Key Features:

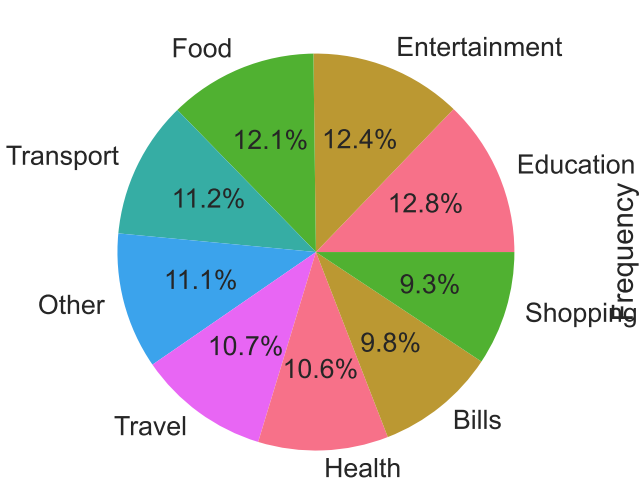
- Automated expense categorization
- Spending trend analysis and forecasting
- Financial anomaly detection
- Interactive visualizations
- Exportable reports and predictions

Data Analysis and Feature Engineering

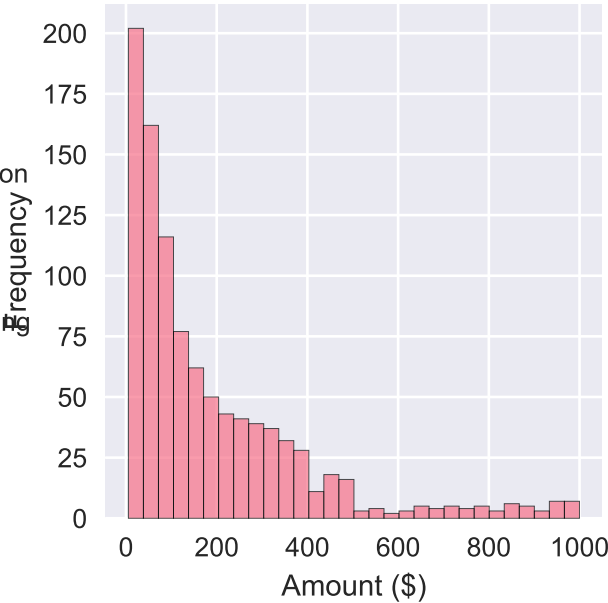
Daily Expenses Over Time



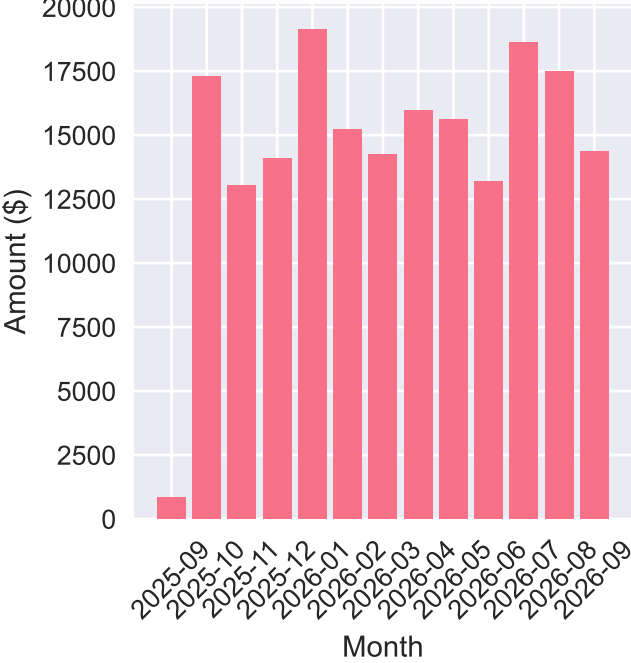
Expense Distribution by Category



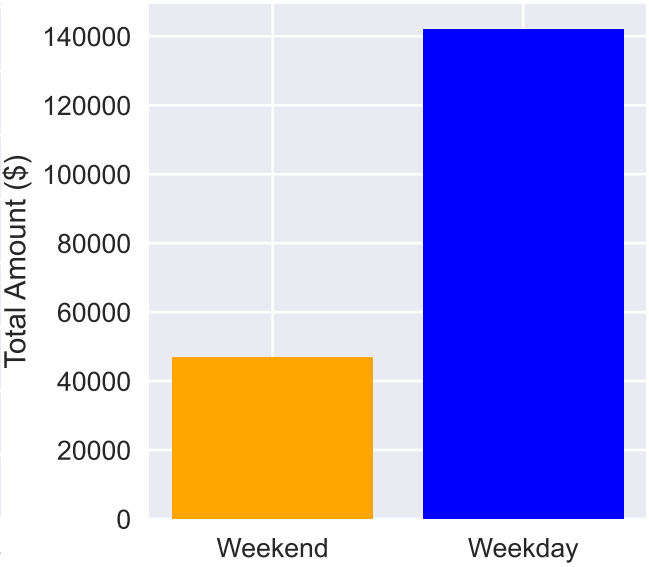
Transaction Amount Distribution



Monthly Spending Trend



Weekend vs Weekday Spending



Feature Engineering Summary:

- Original Features: 8
- Engineered Features: 107
- Total Features: 115

Feature Types:

- Time-based: Hour, day, month, quarter, weekend flags
- Amount-based: Log transform, z-score, rolling statistics
- Text-based: Description length, word count, sentiment
- Categorical: Encoded category, currency, payment method
- Interaction: Amount-time, category-amount combinations

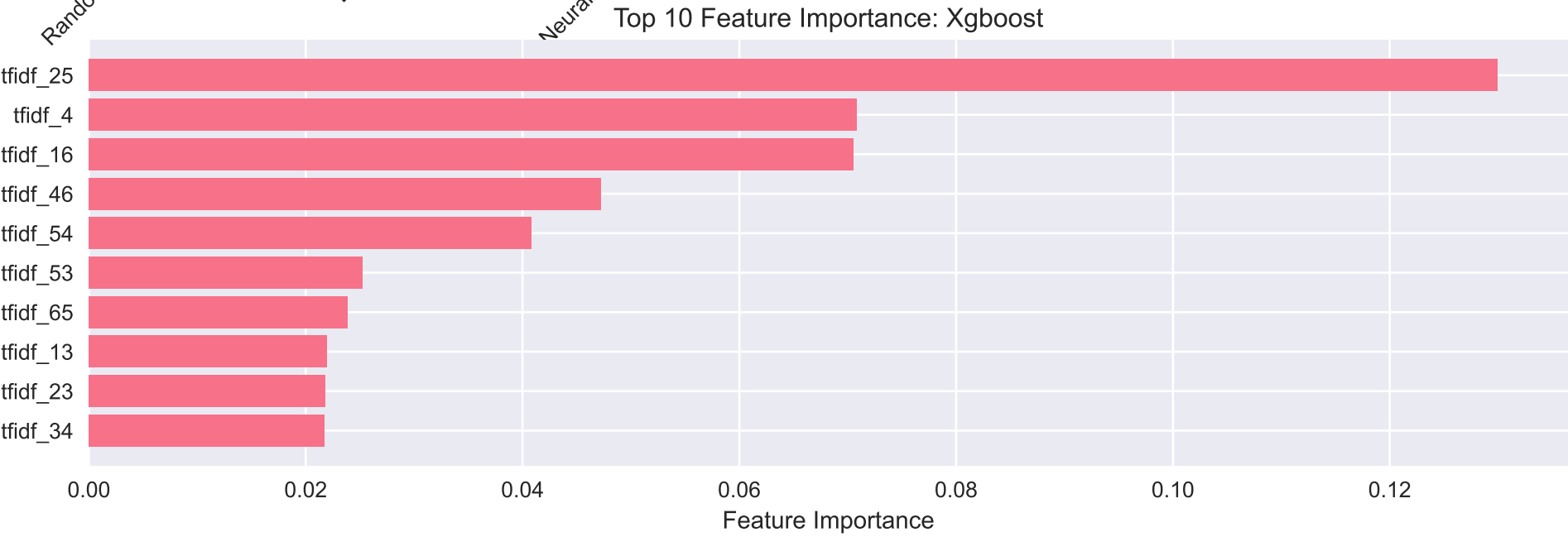
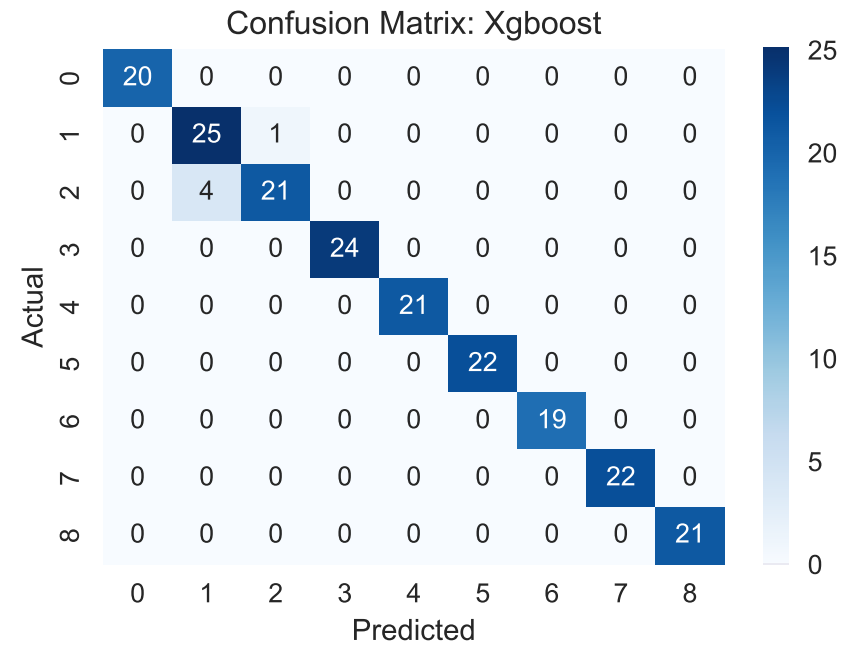
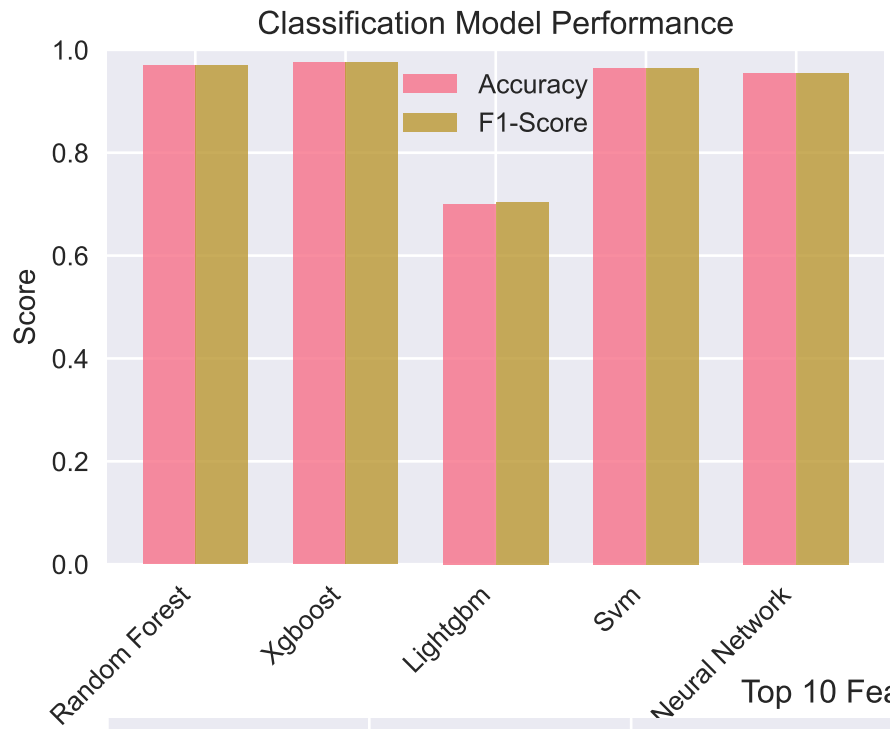
Model Performance Summary

Classification Models		
Model	Accuracy	F1-Score
Random Forest	0.970	0.970
Xgboost	0.975	0.975
Lightgbm	0.700	0.704
Svm	0.965	0.965
Neural Network	0.955	0.955
Regression Models		
Model	R ² Score	RMSE
Linear Regression	0.565	114.176
Ridge	0.565	114.176
Lasso	0.566	114.033
Gradient Boosting	0.570	113.527
Xgboost Regressor	0.473	125.675
Clustering Models		
Model	Clusters	Silhouette Score
Kmeans	5	0.233
Dbscan	0	N/A
Anomaly Detection Models		
Model	Anomaly Rate	Count
Isolation Forest	10.00%	100
One Class Svm	11.10%	111

- Model Performance Metrics:
- Classification: Accuracy (correct predictions), F1-Score (balance of precision & recall)
 - Regression: R² (variance explained), RMSE (prediction error)
 - Clustering: Number of clusters found, Silhouette Score (cluster quality)
 - Anomaly Detection: Percentage of transactions flagged as unusual

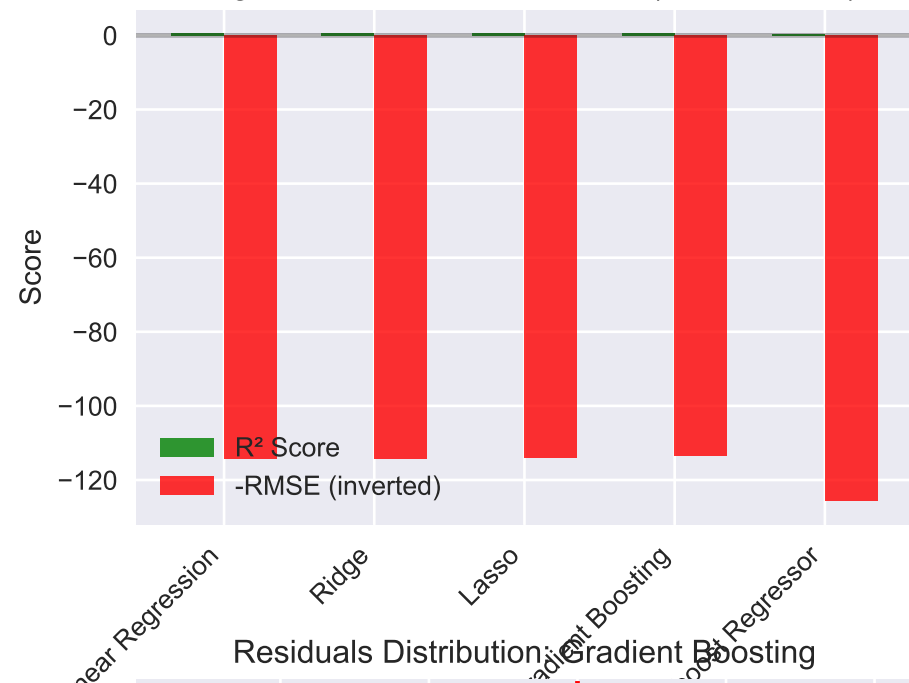
Note: Higher values indicate better performance for all metrics except RMSE (lower is better).

Classification Model Details

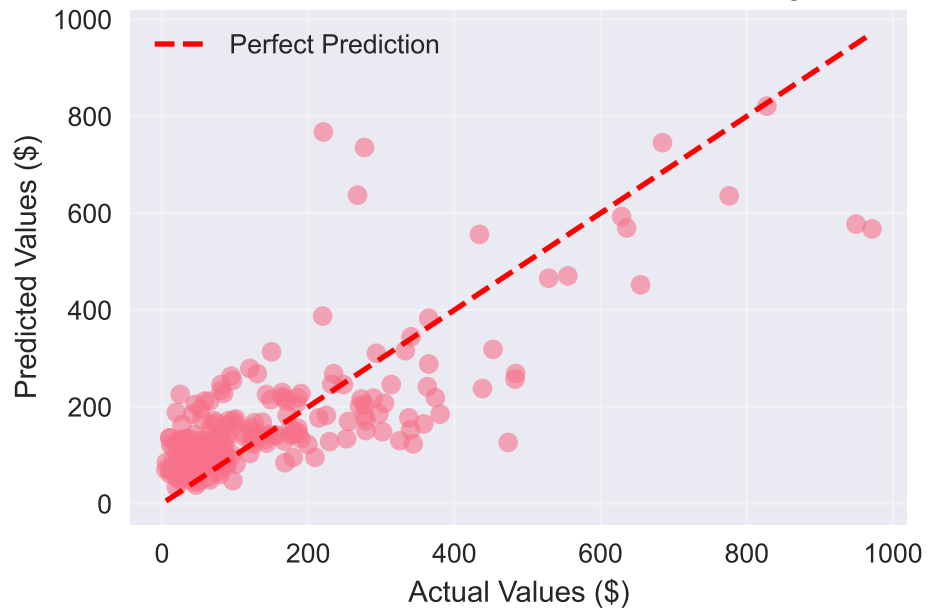


Regression Model Details

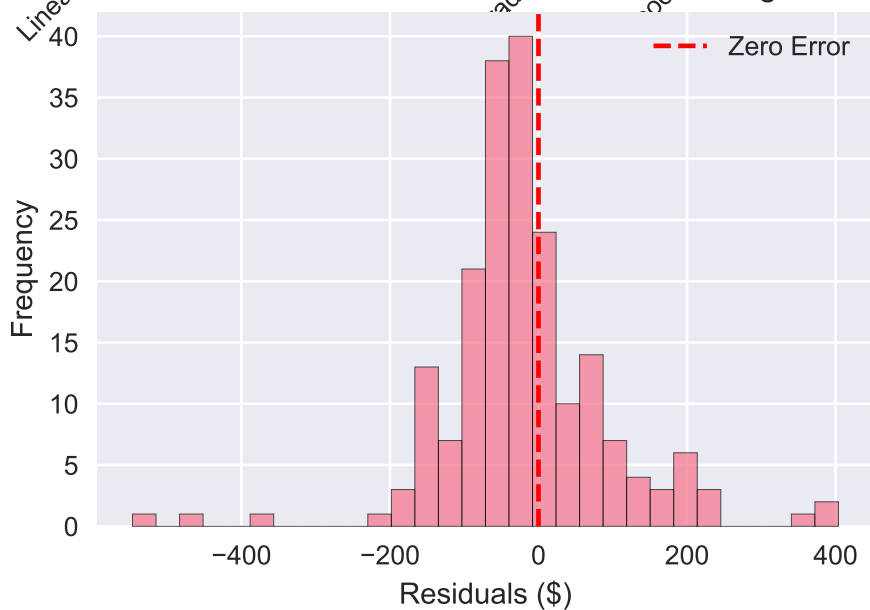
Regression Model Performance (R^2 vs -RMSE)



Predictions vs Actual: Gradient Boosting



Residuals Distribution: Gradient Boosting

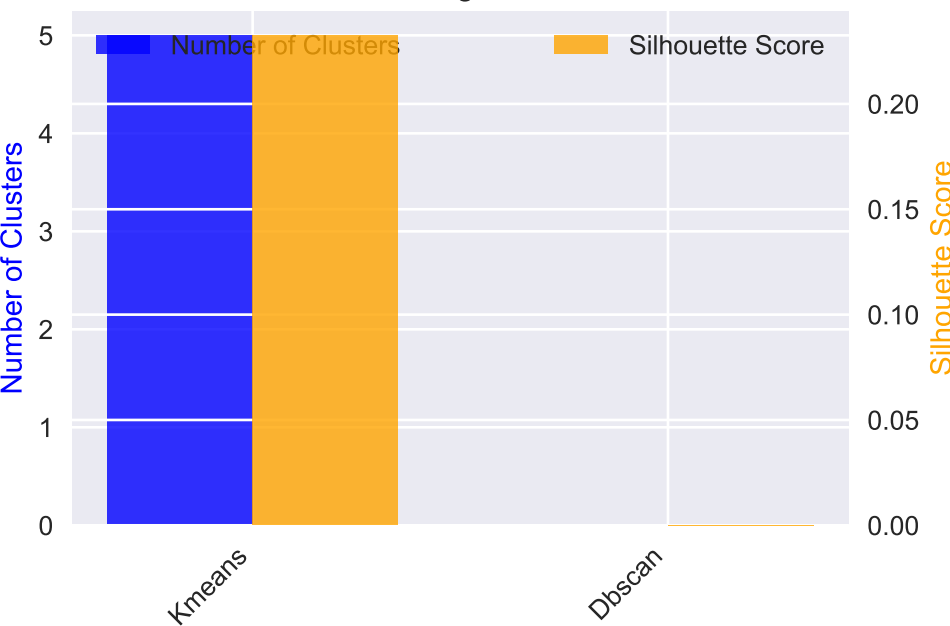


Learning Curves

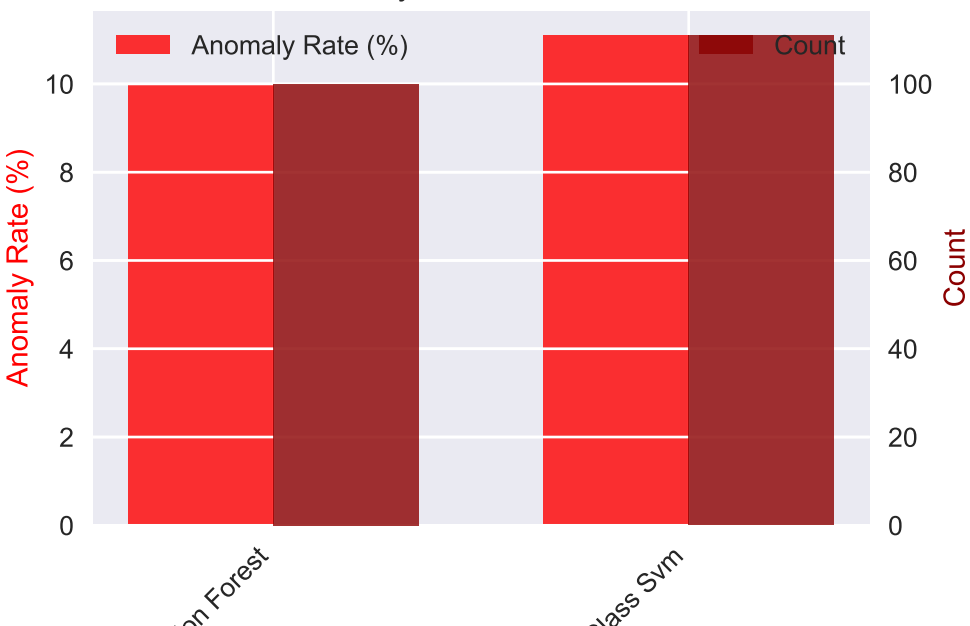


Unsupervised Learning Results

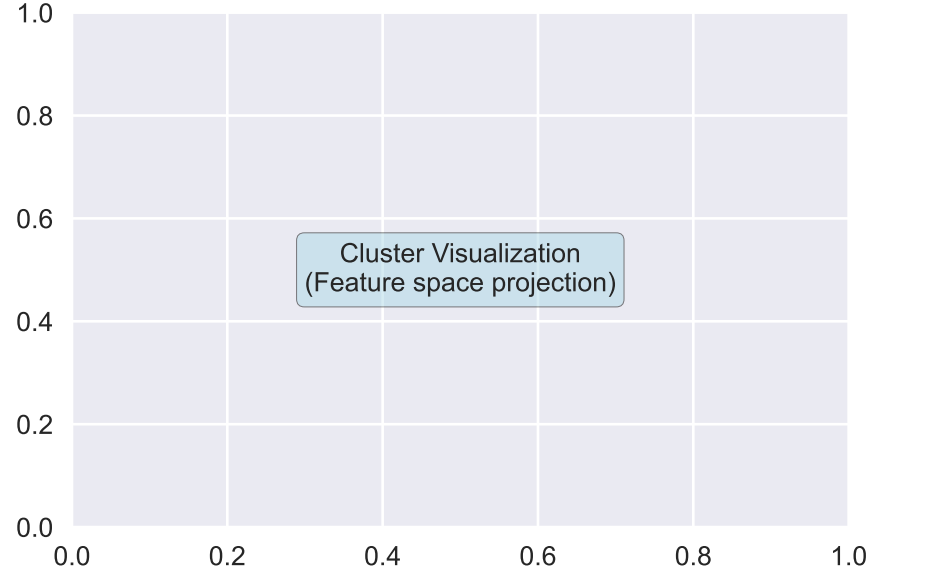
Clustering Results



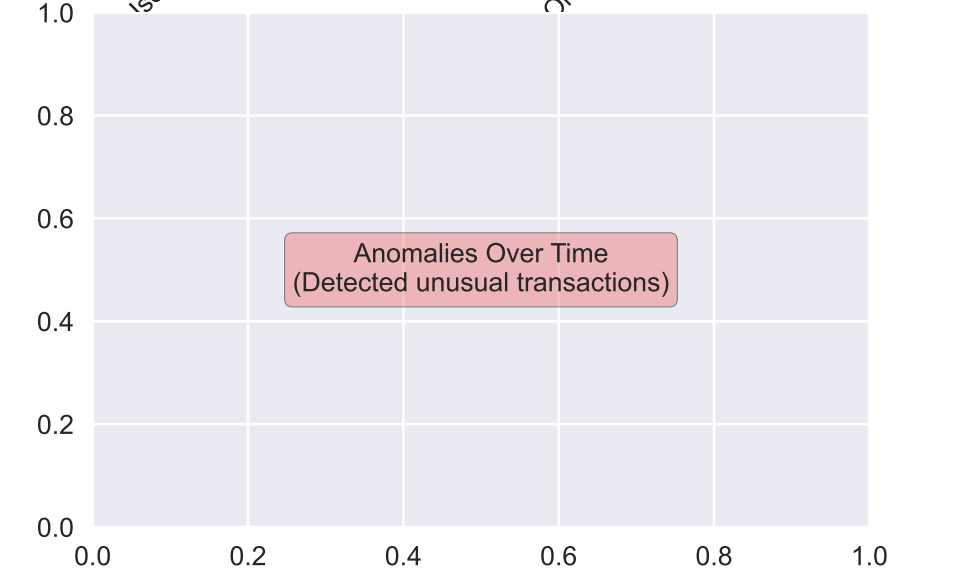
Anomaly Detection Results



Cluster Visualization

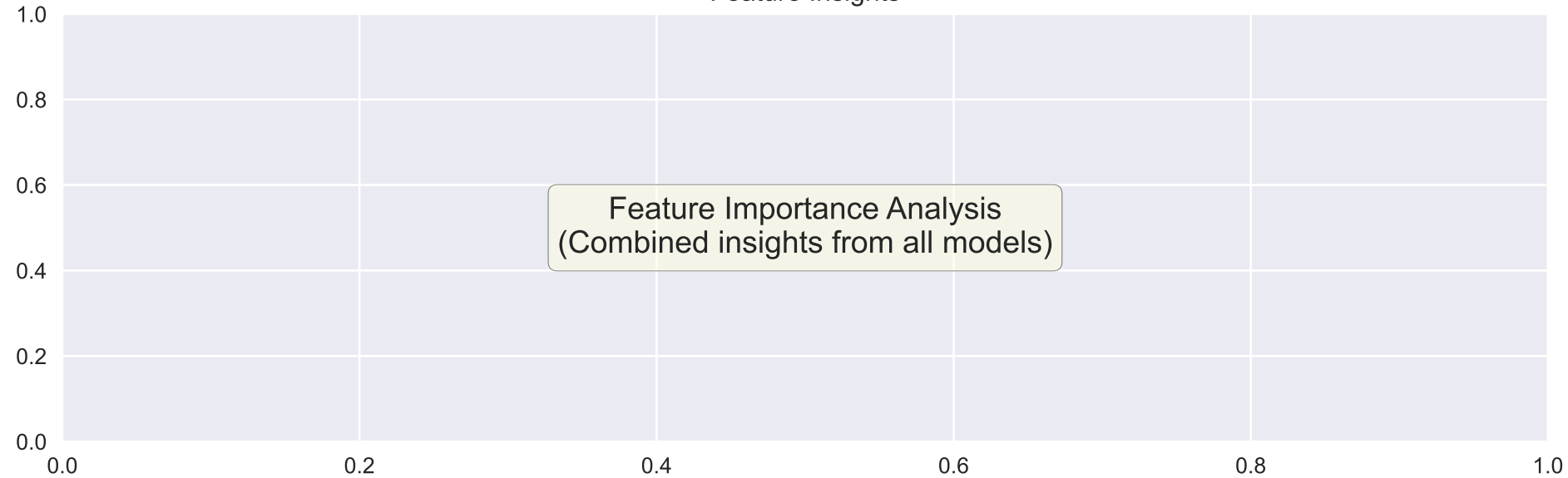


Anomalies Over Time



Insights and Feature Analysis

Feature Insights



Business Insights

Key Spending Insights:

- Categorization accuracy enables automated budgeting
- Prediction models forecast future expenses
- Clustering reveals spending patterns and segments
- Anomaly detection identifies unusual transactions

Business Value:

- Improved financial awareness and control
- Data-driven budgeting decisions
- Early fraud detection
- Personalized financial recommendations

Technical Insights

Technical Implementation:

- Feature engineering: Time, amount, text, categorical features
- Model ensemble: Multiple algorithms for robust predictions
- Evaluation: Comprehensive metrics and cross-validation
- Scalability: Designed for growing datasets

Model Interpretability:

- Feature importance for understanding drivers
- SHAP values available for detailed explanations
- Model cards documenting performance and limitations

Conclusions and Recommendations

EXECUTIVE SUMMARY

Dataset Analysis:

- Processed 1,000 financial transactions
- Total spending: \$189,031.93 over 365 days
- Average transaction: \$189.03
- 9 distinct expense categories identified

Machine Learning Performance:

- Best categorization accuracy: 97.5%
- Best prediction R² score: 0.570
- Multiple algorithms evaluated for robustness

KEY FINDINGS

1. Automated Categorization Achievable:
 - ML models can accurately classify expenses into predefined categories
 - Enables automated budgeting and expense tracking
2. Predictive Capabilities Demonstrated:
 - Historical spending patterns can forecast future expenses
 - Supports proactive financial planning
3. Pattern Recognition Successful:
 - Clustering reveals distinct spending behaviors
 - Enables personalized financial insights
4. Anomaly Detection Functional:
 - Unusual transactions can be flagged for review
 - Enhances financial security and fraud detection

RECOMMENDATIONS

Short-term (0-3 months):

- Deploy categorization model for real-time expense tagging
- Implement monthly spending forecasts
- Set up anomaly alerts for unusual transactions

Medium-term (3-6 months):

- Integrate with banking APIs for automatic data feeds
- Develop web dashboard for interactive visualization
- Add customizable budgeting goals and tracking

Long-term (6+ months):

- Implement deep learning for sequential spending patterns
- Add natural language processing for transaction descriptions
- Create collaborative family/household expense management

TECHNICAL NOTES

- Models trained on historical data - periodic retraining recommended
- Feature engineering pipeline ensures consistent preprocessing
- Evaluation metrics provide unbiased performance estimates
- Configuration-driven design allows easy experimentation